

Development of a Localized E-Learning Platform for Microsoft Office Tool

1st Soe Myint
CSE Department
Parul University

arjunpoudel754@gmail.com

2nd Thu Wai Aung
CSE Department
Parul University
akioyamadar@gmail.com

3rd Myo Thandar
CSE Department
Parul University
assumptamyo@gmail.com

4th Masengo Kinda
CSE Department
Parul University
2303031050797@paruluniversity.ac.in

5th Nidhi Pandey
Assistant Professor
Parul University
nidhi.pandey38091@paruluniversity.ac.in

Abstract – The increase of digital technology has led to an improvement in productivity in various workplaces. As a result, applications such as Microsoft Word, Excel and PowerPoint are widely used in the workplace and in education. The majority of the online courses teaching such skills are conducted in English, which makes it challenging for people who do not know English very well to grasp the lessons. Language barriers can make it impossible for people to learn to operate the programs correctly. The aim of this work is to develop educational means which will help in mastering Microsoft Office skills in different languages. The educational means include lessons in the mother tongue of the students and apply the approaches suitable for their culture and language characteristics. Localization of educational means means not translation of the content from one language to another, but making it appropriate for the language group. The education platform will combine the methods of language training with cultural description.

Keywords – e-learning platform, localization, Microsoft office tools, Django development, Burmese language, French.

In recent years, technology has changed the way we do daily and business tasks. Many digital tools such as Microsoft office(word, PowerPoint, excel) help us to create, present and analyze documents faster and better than traditional methods[10]. Many tasks that used to take hours can now be done in minutes.

Despite Microsoft Office being widely used, there are very few learning resources for people who don't speak English. Most of the popular online tutorials and learning platforms are in English and cannot be translated which makes it harder for non-English speaking countries to depend on those platforms [2].

Countries like Myanmar and other areas in Southeast Asia have many people in rural or semi-urban areas who don't speak English well. As a result, these people can't fully use the available resources to learn digital skills. Many people still use outdated methods for completing tasks that could be done more efficiently with Microsoft Office tools.

Even those with basic knowledge of these tools often only use the most basic functions, missing out on advanced features that could greatly improve their productivity. This is because the learner often find the instructions in English difficult to understand. This research encourages to develop a platform that teaches Microsoft Office tools in the learner's local/native language. This platform promotes

I. INTRODUCTION

systematic lessons, explanations and examples that are culturally relevant to help people with different language background learn more effectively. People tend to use tools such as Google Translate to translate their study material but those tools only translate literally which is not always accurate nor enough. For effective learning, the teaching materials need to be localized rather than translated. The aim is to create a platform that uses the locally spoken languages and slangs as well as culturally relatable. This platform aims to give every local people the feeling of learning from a teacher native to their location, uses their relatable slangs and examples which eventually helps them understand smaller topics clearly. This platform provides locally adapted resources helping people learn these basic but important tools effectively without requiring to learn English or other Global language

II. PROBLEM STATEMENT

People who don't speak English as their first language often face challenges in accessing digital learning resources. Most help materials for productivity software, like Microsoft Office, are in English. This makes it hard for non-English speakers to learn how to use these tools properly. Trying to learn from a foreign-language resource can lead to confusion because of unfamiliar words, instructions, and explanations. As a result, these users may only be able to use the basic features of the software. According to a study by (Ponnusamy et al., 2025), language barriers greatly reduce student engagement and learning effectiveness in digital settings. Many online services rely on automated translation tools like Google Translate to provide content in different languages. However, these tools may not consider the cultural or contextual aspects of the content being translated. Localizing educational content involves more than just translation—it requires adapting the material to fit the syntax, cultural norms, and learning styles of the target audience (P. Sandrini, 1998). P. Sandrini (n.d.) explains that content localization involves adapting digital materials to fit the language and culture of the target audience. This ensures that the learning experience is relevant and effective for all users. Given this point, educational institutions need to have e-learning

platforms specifically designed for learners whose primary language is not the same as English.

As a result, there is an opportunity to create a localized e-learning platform designed to deliver structured training on Microsoft Office tools to learners in their native languages. The development of this type of platform can help to break down language barriers, improve understanding, and help people develop digital literacy skills without having access to appropriate learning resources.

A. Language Barrier in Learning

1) Language plays a critical role in the effectiveness of digital learning platforms, such as MOOCs. When educational content is delivered in a language unfamiliar to learners, comprehension becomes significantly more difficult. According to research, language barrier often reduces a learner's ability to understand instructions[1] in a proper way, causes low interaction with educational systems and decreases learner's engagement with the course content.

2) In many developing countries, learners rely on digital platforms for the acquisition of educational resources. However, most of these learning platform mainly focus on the dominating global languages such as English[2]. Consequently, learners from diverse linguistic backgrounds often struggle to interpret content and may experience reduced motivation and engagement in the learning process.

3) Furthermore, the lack of localized educational resources can limit the effectiveness of digital learning systems. Most people in the urban areas often do not use these digital tools because there are only English resources available to learn from. Because of this, even people with computers and access to internet fail to use these useful tools which can make their daily life and job a lot easier and manageable as well as reduces the paperwork. The challenge faced by these people only emphasizes the importance of developing a localized platform where they can learn immediately and start using these tools in their daily life. Our platform will help those communities with different language background as well as modernize those people who value their culture and language.

4) Therefore, localization is essential in digital learning systems. Unlike simple translation, localization adapts

educational content to match the linguistic style, cultural context, and communication patterns of the target audience. Being able to learn in their native language can help learners understand complex concepts better and apply them in their daily life or work.

TABLE 1
LEARNING EFFICIENCY CHART

Country	English	Local Language
MM Myanmar	50–65%	85–95%
IN India	60–75%	85–95%
JP Japan	55–70%	90–98%
KR South Korea	60–75%	90–98%
CN China	55–70%	90–98%
FR France	65–80%	90–98%

B. Impact of a localized e-learning platform

Localizing educational platforms can greatly enhance digital learning, by allowing learners to access and comprehend educational materials in their languages. This has been demonstrated through research, that providing content to students in their primary language improves access to learning resources and allows them to be more engaged with their digital-learning experience.

Multilingual educational platforms can also provide individualized, customized learning experiences, because the content will be adapted to meet the language preferences, both primary and secondary, as well as to meet the unique learning styles of individual learners.

Multilingual learning platforms also lead to better comprehension and retention of knowledge. Typically, multimedia teaching methods (e.g., video, interactive exercises, localized explanations) make it easier for students to comprehend concepts than do only text-based teaching methods. Research has demonstrated that using multimedia methods for learning results in better comprehension. Many researchers have proved and demonstrated that learning on digital multimedia platform helps learners for better comprehension and retention of the information as compared to classrooms with traditional methods[12] of instructor-to-student teaching.

Localized educational platforms contribute to digital equity by improving access to learning resources for underrepresented communities[14]. By eliminating

language barriers to accessing educational materials, underrepresented communities can access educational resources and build valuable technology skills needed to be successful in today's workplaces. As digital literacy has become essential in modern workplaces, localized education platforms will help diminish the digital divide and create empowered learners.

III. PROPOSED SYSTEM

The proposed e-learning system will be developed using a culturally adaptive and web-based instructional approach that will employ a culturally appropriate and internet based method of delivering instruction in relation to the Microsoft office suite of software. The development of the project will be completed in multiple stages so that language is not a barrier for users learning on the internet.

Stage 1. Requirements Assessment

The requirement assessment will be used to evaluate the requirements of learners who have language-related issues with digital technology. This phase will consist of administering surveys and observing learners who are using English only software applications or online tutorials to assess the challenges they face with the content provided to them. Some items this assessment is focused on include the use of English technical terms, no localized (regional) content on training-support, and the limited availability of structured learning resources for learners.

Stage 2: Content Localization

The content localization phase will be completed using the periodicity of 3 different steps to produce a variety of locations containing the instructional tool(s). The objective of this phase of development is to not simply translate the instructional materials but to also provide explanations, examples, and knowledge of the previous phases of development and formatting based on language characteristics and culture as well. According to P. Sandrini, (1998) localizing is not limited to translation and refers to how digital materials have been adapted to meet the needs of the specific cultures in which the content will be delivered.

Stage 3: Development Of The System as a Website-Based Application

The Web app has been done, and consists of structured modules that provide the necessary courses for Microsoft Office tools, including MS Word, MS Excel, and MS PowerPoint. Each module will contain step-by-step tutorials, examples of how to perform specific tasks, and activation of hands-on practice to increase student-instructor interaction.

Stage 4: Integration of Multimedia for Learning

The system integrates multimedia-based learning materials (e.g., video tutorials, screenshots, guided demonstration) to improve student understanding through the use of both visual and text-based modes of instruction.

Stage 5: User Testing and Evaluation

The last part of the development includes testing the platform with students from various linguistic backgrounds through collecting feedback on ease of use, class effectiveness/value of the class, and overall satisfaction. The information collected through the testing and evaluation procedures will be used to enhance the platform's design and similar types of instructional materials.

IV. METHODOLOGY

In order to provide educational resources in various regions via a distributed system, Django (a web framework) has been selected as the platform upon which the Student Learning Center will be built. In this manner, learners will have access to instructional resources through a web-based application using the Django API.

The frontend and backend of the application will communicate via Django's API to fetch data relevant to the user and display it on the screen in real-time.

The user interface will be designed to allow learning to occur in multiple languages so that users can select their preferred language for each instructional component (tutorials, quizzes, and course materials). Django provides built-in support for internationalization and localization of web applications with regards to language [15], which can be utilized to provide instructional content in multiple languages.

The user interface will have a simple, easily understandable, and intuitive design. This will allow the user to use the Student Learning Center with minimal technical ability, and the interface will have clear navigational aids or aids for assisting users in accessing different sections of the Student Learning Center's user interface.

The application layer is built based on the MVT(Model-View-Template) architecture which supports foundational actions such as user authentication, course management and Progress tracking. Django handles the requests made by the client, interacts with the database, and returns the appropriate response. The localization module is built to accommodate both user and content languages, allowing multilingual educational resources to be delivered in the user's language and culture.

The database is managed by Django's ORM (Object-Relational Mapping) and contains user profile information, course materials, progress information, and assessment results, allowing efficient data storage, retrieval, and management.

A multimedia component is also included to enable the storage of video tutorials, images, and demonstrations to enhance the educational experience through the use of visuals that help illustrate how to use software tools.

Together, these elements work together to create a scalable and efficient e-learning environment capable of providing localized instructional content to a variety of users.

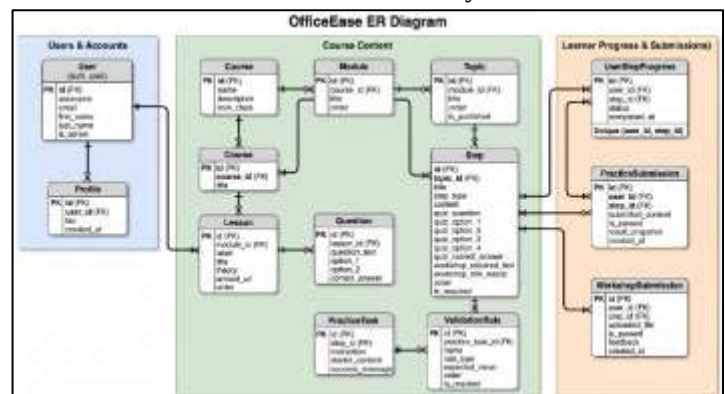


Fig. 1 ER Diagram for the database design

TABLE 2
LITERATURE REVIEW

No.	Author(s)	Year	Research Focus	Method / Approach	Key Findings	Research Gap
1	P. Ponnusamy, A. Gaspar Raja, B. Meenatchi	2025	Language barriers in digital learning platforms	Analysis of digital learning environments	Language barriers significantly affect comprehension and engagement in online learning	Does not propose a localized platform solution
2	A. Al-Azawei, P. Parslow, K. Lundqvist	2016	Challenges of implementing e-learning in developing countries	Survey and qualitative analysis	Infrastructure and language barriers limit adoption of e-learning systems	Limited focus on localized educational platforms
3	Z. Lei	2022	Multilingual web-based learning systems	System design and evaluation	Multilingual interfaces improve accessibility and usability for diverse learners	Focuses on laboratory systems rather than general learning platforms
4	A. Bramantoro, A. Alzahrani, A. Bahaddad	2020	Cloud-based multilingual learning platform	Cloud-based architecture development	Multilingual platforms increase accessibility in smart classrooms	Limited evaluation of language localization impact
5	H. M. Ghadirli, M. Rastgarpour	2013	Multilingual intelligent tutoring systems	Web-based intelligent tutoring system design	Adaptive tutoring systems improve learning efficiency	Does not address localization for general software training
6	N. R. Zhakupov et al.	2024	Development of multilingual online courses	Course development and learner evaluation	Language support improves learner understanding and participation	Focuses mainly on academic courses
7	S. Miranda	2025	AI-assisted multilingual e-learning environments	AI-generated multilingual teaching content	Native language instruction improves engagement and accessibility	Requires advanced AI technologies not accessible in many systems
8	J. Tao	2022	Online language learning challenges	Literature review and analysis	Language difficulties reduce learner participation and completion rates	Does not address practical solutions for digital literacy training

V. FUTURE WORK

While the proposed area of focus for the localized e-learning platform will be on training with the Microsoft Office suite of applications, there are many opportunities for future enhancements and development.

One avenue for expanding the platform would be to include other software applications, along with a variety of digital skills training programs. Courses could also address programming languages, data management software, web development platforms, etc. The broader range of courses would enable the platform to provide services to a wider audience, while promoting digital literacy across a variety of disciplines.

Another potential area of future research would be the application of AI technologies to develop personalized learning experiences for users. AI recommendation systems

could analyse user patterns, and develop custom-made learning paths for users with specific needs and skill levels. In addition, future enhancements might include the creation of mobile apps that would give users easier access to the platform. Many users currently use only mobile phones, so the mobile app would help to expand the reach of digital learning to areas where there may be limited access to desktops.

Finally, future research may explore the inclusion of voice-activated learning systems, along with the implementation of voice recognition technology to help users that are unable to read traditional printed instructions. Such capabilities would help to improve access and inclusion into digital learning environments.

VI. CONCLUSION

This research demonstrates how localization impacts digital learning platforms. Not being able to read or understand the globalized contents are usually the biggest obstacles faced by those looking to acquire the necessary skills through an online learning system that provides digital skill development, when teaching materials are presented in a language other than their native tongue. This may make it difficult to comprehend the content presented and effectively use the digital tools as needed.

Localized e-learning platforms offer a potential solution to this situation. E-learning platforms that provide instructional materials in learners' native languages will help improve the level of comprehension and engagement of learners while increasing the accessibility of the e-learning system to individuals learning through this mode of education.

In addition to the creation of localized e-learning platforms, the creation of instructional materials using multimedia resources (video/audio/experiential learning) and employing culturally relevant examples as well as designing user-friendly interfaces, has the potential to significantly improve the quality of education delivered by localized e-learning platforms.

The development of a localized e-learning platform to deliver Microsoft Office training will make it possible for all learners, regardless of their native language, to access digital education and become digitally literate. Furthermore, by addressing the impact of language on learning and increasing the level of digital literacy, localized e-learning platforms will provide individuals with the means to be engaged and productive in the global digital economy, while fostering greater educational inclusion.

REFERENCES

- [1] P. Ponnusamy, A. Gaspar Raja, and B. Meenatchi, *Barriers on English Language Learning Through Digital Platforms and its Impact on Primary School Students*, 2025.
- [2] A. Azawei, P. Parslow, and K. Lundqvist, *Barriers and Opportunities of E-Learning Implementation in Developing Countries*, International Review of Research in Open and Distributed Learning, 2016.
- [3] Z. Lei, "A Web-Based Multi-Language System for Remote and Virtual Laboratories," Heliyon, 2022.
- [4] A. Bramantoro, A. Alzahrani, and A. Bahaddad, "Cloud-Based Learning Service Platform for Multilingual Smart Class," International Journal of Advanced and Applied Sciences, 2020.
- [5] H. M. Ghadirli and M. Rastgarpour, "A Web-Based Multilingual Intelligent Tutor System," International Journal of Computer Applications, 2013.
- [6] N. R. Zhakupov et al., "Development of a Multilingual Online Course with Language Support," International Journal of Evaluation and Research in Education, 2024.
- [7] S. Miranda, "Enhancing Multilingual E-Learning Experiences Using AI-Generated Virtual Speakers," Information Journal, 2025.
- [8] J. Tao, "Teaching and Learning Languages Online: Challenges and Opportunities," 2022.
- [9] A. Mandala, "Distance Learning and Multilingual Education," International Journal of Curriculum and Education, 2025.
- [10] R. Soni, "Impact of Digital Learning Platforms on Student Engagement and Performance," International Journal of Creative Research and Management, 2024.
- [11] M. H. Miraz, M. Ali, and P. S. Excell, "Cross-Cultural Usability Issues in E/M-Learning," International Journal of Computer Applications, vol. 179, no. 21, pp. 1–5, 2018.
- [12] R. P. Díaz-Redondo, A. Fernández-Vilas, M. Pazos-Arias, and M. Ramos-Cabrer, "Integrating Micro-Learning Content in Traditional E-Learning Platforms," IEEE Access, vol. 11, pp. 23456–23468, 2023.
- [13] S. Thuseethan, S. Achchuthan, and S. Kuhanesan, "Usability Evaluation of Learning Management Systems," International Journal of Computer Applications, vol. 89, no. 6, pp. 10–16, 2014.
- [14] M. Ally and Y. Beldarrain, "Transforming Education with Mobile Learning," International Review of Research in Open and Distributed Learning, vol. 20, no. 2, pp. 1–15, 2019.
- [15] P. Sandrini, "Localization and Translation," 1998.